

BUSINESS UNDERSTANDING REPORT: HEART DISEASE PREDICTION



(Jeevi Academy image)

Business Problem

Cardiovascular disease is one of the leading causes of death globally, and early detection significantly improves patient outcomes and reduces treatment costs. In a clinical or healthcare-insurance setting, identifying patients at high risk of heart disease before symptoms become severe allows for earlier intervention, more efficient allocation of medical resources, and better long-term health outcomes.

Currently, risk assessment often relies on individual clinician judgement and standard diagnostic thresholds, which can miss patients who don't present obvious symptoms until the disease has progressed.

Project Objective

The objective of this project is to prepare a machine-learning-ready dataset that can support the development of a predictive model capable of estimating a patient's likelihood of having heart disease based on clinical and demographic attributes.

Before any model is built, the raw data must be cleaned, transformed, encoded, scaled, and evaluated for feature relevance, ensuring the eventual model is trained on high-quality, well-structured data.

Target Variable

The target variable is **heart disease**, a binary indicator where:

- 1 = the patient has heart disease.
- 0 = the patient does not have heart disease.

This makes the underlying problem a binary classification task. The dataset includes predictor variables such as age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar and etc., all of which are clinically relevant to cardiovascular risk assessment.

Importance of Predictive Modelling

- **Consistency:** a model applies the same criteria uniformly across every patient, reducing variability introduced by individual clinician interpretation.
- **Early risk alert or flagging:** a model can quantify risk probability rather than relying on binary pass/fail thresholds on individual metrics, catching borderline cases a simple rule might miss.
- **Cost reduction:** Lowers expensive hospital readmissions and avoids emergency room visits.

- **Improve Outcomes:** Reduces mortality rates by stopping chronic diseases from progressing.

Expected Business Impact

If deployed responsibly and validated by clinical professionals, a predictive model built on this dataset could support:

- **Earlier interventions:** reducing the likelihood of acute cardiac events and their associated treatment costs.
- **More efficient resource allocation:** allowing healthcare providers to prioritize high-risk patients for further testing or specialist referral.
- **Better-informed patients:** giving individuals a clearer, quantified sense of their cardiovascular risk.